Richard Haburcak

Curriculum Vitae

Education

2018–2023 **Ph.D. in Mathematics**, Dartmouth College, Hanover, New Hampshire
Advisor: Asher Auel
Thesis: Brill–Noether Theory via K3 Surfaces

2014–2018 **M.A. in Chemistry**, Brandeis University, Waltham, Massachusetts
Advisor: Bing Xu
Thesis: Ligand-Receptor Interactions Modulate Self-Assembly of Small Molecules

2014–2018 **B.A. in Mathematics and Chemistry**, Brandeis University, Waltham, Massachusetts

Appointments

2024-Present **RTG Visiting Assistant Professor**, The Ohio State University, Columbus, Ohio

Fall 2026 **Semester Program Participant**, ICERM Semester Program on Computations on K3 surfaces and Related Varieties, Providence, Rhode Island

Spring 2024 **Postdoctoral Fellow**, Max Planck Institute for Mathematics, Bonn, Germany

Fall 2023 **Junior Trimester Program**, Hausdorff Research Institute for Mathematics, Bonn, Germany

2023-2024 **Visiting Scholar**, Dartmouth College, Hanover, New Hampshire

Awards and Grants

2025-2027 **AMS-Simons Travel Grant**, *$6,000*

2023 **Outstanding Graduate Teacher**, *Dartmouth College*, student nominated award

2018 **Shapiro Prize in Mathematics**, *Brandeis University*

2017 **Goldwater Scholarship Honorable Mention**

2016 **Science Hall of Fame**, *Brandeis University*

Publications and Preprints

Preprints

Link *Brill–Noether loci in genus* ≤ 12, Richard Haburcak. 2025. Submitted.

Link *Distinguishing Brill–Noether loci*, Asher Auel, Richard Haburcak, and Andreas Leopold Knutsen. 2024. Submitted.

Publications

Link *Some reducible and irreducible Brill–Noether loci*, Richard Haburcak and Montserrat Teixidor i Bigas. Math. Z. **312**, 102 (2026).

haburcak.1@osu.edu

Link *Maximal Brill–Noether loci via K3 surfaces*, Asher Auel and Richard Haburcak. International Mathematics Research Notices (IMRN), **2025**(20), 2025.

Link *Maximal Brill–Noether loci via degenerations and double covers*, Andrei Bud and Richard Haburcak. 2024. Accepted to Ann. Sc. Norm. Super. Pisa Cl. Sci. (5).

Link *Holomorphic forms and non-tautological cycles on moduli spaces of curves*, Veronica Arena, Samir Canning, Emily Clader, Richard Haburcak, Amy Q. Li, Siao Chi Mok, and Carolina Tamborini. Sel. Math. New Ser. 31, 47 (2025).

Link *Maximal Brill–Noether loci via the gonality stratification*, Asher Auel, Richard Haburcak, and Hannah Larson. Accepted to Michigan Math. J.

Link *Curves on Brill–Noether special K3 surfaces*, Richard Haburcak. Math. Nachr. 297 (2024), 4479–4509.

Previous Chemistry Work

Link *Ligand–Receptor Interaction Modulates the Energy Landscape of Enzyme-Instructed Self-Assembly of Small Molecules*, Richard Haburcak , Junfeng Shi, Xuewen Du, Dan Yuan, and Bing Xu*, *J. Am. Chem. Soc.* **2016**, 138, 47, 15397–15404.

Link *Nanonets Collect Cancer Secretome from Pericellular Space*, Rong Zhou, Yi Kuang, Jie Zhou, Xuewen Du, Jie Li, Junfeng Shi, Richard Haburcak, Bing Xu*, *PLOS One* **2016**, 11, e0154126.

Link *Enzymatic Dissolution of Biocomposite Solids Consisting of Phosphopeptides to Form Supramolecular Hydrogels*, Junfeng Shi, Dan Yuan, Richard Haburcak, Qiang Zhang, Dr. Chao Zhao, Prof. Dr. Xixiang Zhang, PRof. Dr. Bing Xu*, *Chem. Eur. J.* **2015**, 21, 18047-18051

Link *Supramolecular Detoxification of Neurotoxic Nanofibrils of Small Molecules via Morphological Switch*, Junfeng Shi, Xuewen Du, Dan Yuan, Richard Haburcak, Ning Zhou, and Bing Xu*, *Bioconjugate Chem.* **2015**, 26, 9, 1879–1883.

Link *Enzyme transformation to modulate the ligand–receptor interactions between small molecules*, Junfeng Shi, Xuewen Du, Dan Yuan, Richard Haburcak, Dongdong Wu, Ning Zhou, and Bing Xu*, *Chem. Commun.* **2015**, 51, 4899-4901

Link *Ligand–Receptor Interaction Catalyzes the Aggregation of Small Molecules To Induce Cell Necroptosis*, Junfeng Shi, Xuewen Du, Yibing Huang, Jie Zhou, Dan Yuan, Dongdong Wu, Ye Zhang, Richard Haburcak, Irving R. Epstein*, and Bing Xu*, *J. Am. Chem. Soc.* **2015**, 137, 26–29.

Teaching experience

The Ohio State University, Instructor of record

Spring 2026 **Abstract Algebra 1**, *Math 4580*

Fall 2025 **Abstract Algebra 2**, *Math 4581*

Spring 2025 **Introduction to ODEs**, *Math 2255*

Fall 2024 **Introduction to ODEs**, *Math 2255*

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Dartmouth College, Instructor of Record

Winter 2023 **Multivariable Calculus**, *Math 13*

Winter 2022 **Calculus**, *Math 3*

Fall 2020 **Introduction to Calculus**, *Math 1*

Teaching Assistant

Winter 2020 **Graduate Algebra**, *Math 101*, Dartmouth College

Fall 2019 **Calculus of Functions of one and Several Variables**, *Math 8*, Dartmouth College

Spring 2019 **Differential Equations**, *Math 23*, Dartmouth College

Fall 2018 **Advanced Multivariable Calculus**, *Math 13*, Dartmouth College

2015–2017 **Quantum Chemistry**, *Chem 142*, Brandeis University, 3 times

2017–2018 **Thermodynamics**, *Chem 141*, Brandeis University, 2 times

Talks

Invited Conference Talks

April 2026 **K3 surfaces and Brill–Noether theory**, *AMS Spring Central Sectional on Classical questions through the lens of birational algebraic geometry*, North Dakota State University, Fargo, North Dakota

Jan 2026 **Refined Brill–Noether theory**, *AMS Special Session on Combinatorics and moduli spaces*, JMM 2026, Washington, D.C.

Oct 2025 **Brill–Noether loci**, *Boston Algebraic Geometry Day*, Tufts University, Medford, Massachusetts

June 2025 **Limit linear series, theta characteristics, and Brill–Noether loci**, *3rd Workshop of the Kentucky–Ohio ALgebra Alliance (KOALA 2025)*, The Ohio state University, Columbus, Ohio

Jan 2024 **The Maximum Gonality in a Brill–Noether Locus**, *AMS Special Session on Combinatorial Perspectives on Algebraic Curves and their Moduli*, JMM 2024, San Francisco, California

April 2023 **Brill–Noether special cubic fourfolds of discriminant** 26, *AMS Spring Central Sectional on Brauer Groups in Algebraic Geometry and Arithmetic*, University of Cincinnati, Cincinnati, Ohio

Oct 2022 **Brill–Noether Special K3 Surfaces**, *AMS Fall Southeastern Sectional on Geometry and Arithmetic of Hyperkähler Manifolds*, University of Tennessee, Chattanooga, Tennessee

Invited Seminar Talks

Oct 2026 **Brill–Noether special K3 surfaces**, *Algebraic Geometry Seminar*, Brown University, Providence, Rhode Island

Oct 2026 **K3 surfaces and Brill–Noether theory**, *Algebra Seminar*, Auburn University, Auburn, Alabama

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Oct 2026 **K3 surfaces and Brill–Noether theory**, *Algebra Seminar*, Georgia Tech, Atlanta, Georgia

Oct 2026 **K3 surfaces and Brill–Noether theory**, *Geometry and Topology Seminar*, Virginia Tech, Blacksburg, Virginia

Oct 2025 **Brill–Noether loci in low genus**, *Algebra Seminar*, University of Kentucky, Lexington, Kentucky

Sept 2025 **Special Algebraic Curves**, *Ohio State University Math Welcome Seminar*, The Ohio State University, Columbus, Ohio

March 2025 **Brill–Noether loci in low genus**, *Algebra, Geometry, and Number Theory Seminar*, Tufts University, Medford, Massachusetts

March 2025 **Brill–Noether loci in low genus**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

Jan 2025 **Non-tautological Cycles on Moduli Spaces of Smooth Curves**, *Arithmetic Geometry Seminar*, The Ohio State University, Columbus, Ohio

Aug 2024 **Distinguishing Brill–Noether loci**, *Algebraic Geometry Seminar*, The Ohio State University, Columbus, Ohio

June 2024 **Distinguishing Brill–Noether loci**, *Seminar Algebraic Geometry*, MPIM Bonn, Bonn, Germany

May 2024 **Maximal Brill–Noether loci**, *Oberseminar*, MPIM Bonn, Bonn, Germany

April 2024 **(Algebraic) Surfaces, lattice polarized K3 surfaces, and Brill–Noether special K3s**, *Geometry and Topology Seminar*, University of Iowa, Iowa City, Iowa

Dec 2023 **Maximal Brill–Noether loci**, *Geometry Seminar*, Università Roma Tre, Rome, Italy

Nov 2023 **Maximal Brill–Noether loci**, *Research Seminar Algebraic Geometry*, Leibniz Universität Hannover, Hannover, Germany

Oct 2022 **Maximal Brill–Noether loci via K3 Surfaces**, *Algebra Seminar*, University of Kentucky, Lexington, Kentucky

Sept 2022 **Maximal Brill–Noether loci via K3 Surfaces**, *Algebra Seminar*, Brown University, Providence, Rhode Island

April 2022 **Maximal Brill–Noether Loci via Lifting Line Bundles**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

March 2021 **Brill-Noether Special K3 Surfaces**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

Expository and Reading Seminar Talks

March 2026 **(Moduli stack of) Vector Bundles on P1 (cont.)**, *RTG learning seminar on Affine Grassmannian*, The Ohio State University, Columbus, Ohio

Feb 2026 **(Moduli stack of) Vector bundles on P1**, *RTG learning seminar on Affine Grassmannian*, The Ohio State University, Columbus, Ohio

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Oct 2025 **Degeneracy Loci**, *RTG learning seminar on Affine Grassmannian*, The Ohio State University, Columbus, Ohio

Nov 2023 **Torelli theorems for cubic threefolds**, *STReTCH seminar*, Hausdorff Institute for Mathematics, Bonn, Germany

April 2023 **Why are word problems hard? (Introduction to the word problem for groups)**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Oct 2022 **Galois Cohomology and Local Class Field Theory**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

Oct 2022 **All Triangles Form a Triangle**, *Undergraduate Math Club*, University of Kentucky, Lexington, Kentucky

Sept 2022 **Triangles. Boundaries. And...? (Introduction to moduli spaces of triangles)**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

July 2022 **Ètale Cohomology**, *Brauer Group Reading Seminar*, Dartmouth College, Hanover, New Hampshire

July 2022 **Group and Galois Cohomology**, *Brauer Group Reading Seminar*, Dartmouth College, Hanover, New Hampshire

April 2022 **Sheaves on Graphs**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Jan 2022 **Introduction to Algebraic Surfaces I**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

Oct 2021 **Categories, Particle Physics, and Yoneda**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

May 2021 **Finite groups, étale groups, Cartier duality**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

April 2021 **Projective Space and Lines**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

March 2021 **Nekovář's p-adic Heights**, *Reading Seminar in Number Theory*, Dartmouth College, Hanover, New Hampshire

Jan 2021 **Sheafy Vector Bundles and Chern Classes**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Nov 2020 **Multiplying Divisors**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Jan 2020 **Maximal Spectra and Why Continuous Functions are Important**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

Sept 2019 **Semisimplicity of Finite Group Rings**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

April 2019 **Adding and Putting Rings on It (Abelian Categories)**, *Graduate Student Seminar*, Dartmouth College, Hanover, New Hampshire

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Service and Outreach

Fall 2026 - Spring 2027 **Mentor**, *OSU Cycle (undergraduate reading and research)*, The Ohio State University, Columbus, Ohio

Fall 2025 - Spring 2026 **Mentor**, *OSU Cycle (undergraduate reading and research)*, The Ohio State University, Columbus, Ohio

Spring 2023 **Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Winter 2023 **Co-organizer and Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Spring 2022 **Organizer and Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Winter 2022 **Organizer**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Winter 2021 **Organizer and Mentor**, *Directed Reading Program*, Dartmouth College, Hanover, New Hampshire

Spring 2019 **Co-organizer**, *Math table at Dartmouth Science Day*, Dartmouth College, Hanover, New Hampshire

Conference Organizing

Spring 2026 **Co-organizer**, *AMS Special Session on Moduli and Syzygies of Algebraic Curves, as part of AMS Spring Eastern Sectional*, Boston College, Boston, Massachusetts

Fall 2023 **Co-organizer**, *School on K3 surfaces, hyperkähler manifolds, and cubic fourfolds, as part of a Junior Trimester Program*, Hausdorff Research Institute for Mathematics, Bonn, Germany

Seminar Organizing

Fall 2026 **Co-organizer**, *ICERM Semester Program, Graduate and Postdoc seminar*, ICERM, Brown University, Providence, Rhode Island

2025-Present **Co-organizer**, *Ohio State University Algebraic Geometry Seminar*, The Ohio State University, Columbus, Ohio

Summer 2022 **Co-organizer**, *Brauer Group Reading Seminar*, Dartmouth College, Hanover, New Hampshire

2021-2022 **Co-organizer**, *Dartmouth Algebra and Number Theory Seminar*, Dartmouth College, Hanover, New Hampshire

Conferences and Workshops Attended

Nov 2026 **AGNES 2026**, Brown University, Providence, Rhode Island

Sept-Dec 2026 **Semester Program on Computations on K3 Surfaces and Related Varieties**, ICERM, Providence, Rhode Island

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Aug 2026	**Workshop on Moduli of curves, abelian varieties and K3 surfaces**, Erdős Center, Rènyi Institute, Budapest, Hugary
April 2026	**Affine Combinatorics and Quantum K-Theory, OSU RTG retreat 2026**, The Ohio State University, Columbus, Ohio
April 2026	**KOALA 2026**, University of Kentucky, Lexington, Kentucky
April 2026	**AMS Spring Central Sectional**, North Dakota State University, Fargo, North Dakota
Mar 2026	**AMS Spring Eastern Sectional**, Boston College,Boston, Massachusetts
Jan 2026	**VBAC 2026**, Universidad Autónoma de Yucatán (UADY), Mérida, Yucatán, Mexico
Jan 2026	**JMM 2026**, Washington, D.C.
Nov 2025	**Algebra, Geometry and Combinatorics Day (AlGeCom) XXVI**, The Ohio State University, Columbus, Ohio
Oct 2025	**Boston Algebraic Geometry Day** , Tufts University, Medford, Massachusetts
Sept 2025	**Workshop on Combinatorics and Geometry of Moduli of Curves**, University of Illinois Urbana-Champaign, Champaign, Illinois
July 2025	**Summer Research Institute in Algebraic Geometry**, Colorado State, Fort Collins, Colorado
July 2025	**Bootcamp preceding Summer Research Institute in Algebraic Geometry**, Colorado State, Fort Collins, Colorado
June 2025	**3rd Workshop of the Kentucky-Ohio ALgebra Alliance (KOALA 2025)**, The Ohio State University, Columbus, Ohio
May 2025	**Arithmetic, K-theory, and Algebraic Cycles**, The Ohio State University, Columbus, Ohio
March 2025	**MAGNTS 2025**, The Ohio State University, Columbus, Ohio
March 2025	**Georgia Algebraic Geometry Symposium 2025**, Univerisyt of Georgia, Athena, Georgia
Nov 2024	**AGNES 2024**, Dartmouth College, Hanover, New Hampshire
Feb 2024	**A panorama of moduli spaces**, Goethe University Frankfurt am Main, Frankfurt, Germany
Jan 2024	**JMM 2024**, San Francisco, California
Oct 2023	**Vector bundles and combinatorial algebraic geometry**, Goethe University Frankfurt am Main, Frankfurt, Germany
Sept 2023	**School on K3 surfaces, hyperkähler manifolds, and cubic fourfolds**, Hausdorff Research Institute for Mathematics, Bonn, Germany
Aug 2023	**VBAC 2023**, Universität Duisburg-Essen, Essen, Germany
July 2023	**AGNES Summer School on Intersection Theory on Moduli Spaces**, Brown University, Providence, Rhode Island
July 2023	**GAeL XXX**, University of Warwick, UK

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April 2023 **AGNES 2023**, Stony Brook University, Stony Brook, New York

April 2023 **AMS Spring Central Sectional**, University of Cincinnati, Cincinnati, Ohio

Jan 2023 **JMM 2023**, Boston, Massachusetts

Dec 2022 **Birational Complexity of Algebraic Varieties 2022**, Simons Center, Stony Brook, New York

Dec 2022 **RTG/SCGP Workshop on Birational Complexity of Algebraic Varieties 2022**, Simons Center, Stony Brook, New York

Nov 2022 **AGNES 2022**, UMass Amherst, Amherst, Massachusetts

Oct 2022 **AMS Fall Southeastern Sectional**, University of Tennessee, Chattanooga, Tennessee

Oct 2022 **AMS Fall Eastern Sectional**, UMass Amherst, Amherst, Massachusetts

Sept 2022 **BATMOBILE 2022**, Amherst College, Amherst, Massachusetts

Aug 2022 **AGNES Summer School on Higher Dimensional Moduli**, Brown University, Providence, Rhode Island

July 2022 **VBAC 2022**, University of Warwick, UK

May 2022 **AGNES 2022**, Rutgers University, New Brunswick, New Jersey

May 2022 **BATMOBILE 2022**, Brown University, Providence, Rhode Island

Oct 2021 **AGNES 2021**, Boston College, virtual

Jan 2021 **JMM 2021**, virtual

Oct 2020 **AGNES 2020**, Stony Brook, virtual

Referee

Semigroup Forum
Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg
Algebraic Geometry
Mathematische Zeitschrift

Press

Math Directed Reading Program Fosters Mentorship and Inclusion

Memberships

American Mathematical Society (AMS).

Languages

English: native speaker. **Slovak:** fluent. **Spanish:** intermediate.

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